Habib University

Course Title: Cryptography and Network Security (EE-426 / CE-442)

Instructors: Farhan Khan

Assignment No. 02 Solution

Release Date: March 31st, 2026	**Due by:** April 6th, 2026 11:59 PM
Total points: 100	**Points obtained:**

Student Name:	**Student ID:**	**Section:**

Purpose:

The purpose of this assignment is to help you explain the concepts behind several symmetric-key and asymmetric-key cryptosystems.

Instructions:

1. This assignment should be done individually.
2. All questions should be answered in **black ink only**. (Extra sheets can be used)
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor/TA.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work will result in a reduction of your points.

Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	✓
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	

P#	Questions	Pts
1	Apply each of the following AES transformations on the following two plaintexts that differ only in the first bit assuming the use of same cipher key. P1: $(0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0001)_{16}$ P2: $(8000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0001)_{16}$ a. SubBytes b. ShiftRows c. MixColumns d. AddRoundKey (with the same round keys of your choice) [Hint: You can use a round key with all zeros here] Find the number of bits changed after each transformation assuming each transformation is applied one after the other in order.	20

Answer:

- **Initial States**

P_1

$$\begin{bmatrix} 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 01 \end{bmatrix}$$

P_2

$$\begin{bmatrix} 80 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 01 \end{bmatrix}$$

a) SubBytes

Using AES S-box:

- $S(00) = 63$
- $S(80) = CD$
- $S(01) = 7C$

After SubBytes — P_1

$$\begin{bmatrix} 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 7C \end{bmatrix}$$

After SubBytes — P_2

$$\begin{bmatrix} CD & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 7C \end{bmatrix}$$

b) ShiftRows

Shift pattern:

- Row 0 → no shift
- Row 1 → left 1
- Row 2 → left 2
- Row 3 → left 3

After ShiftRows — P_1

$$\begin{bmatrix} 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 7C & 63 & 63 & 63 \end{bmatrix}$$

After ShiftRows — P_2

$$\begin{bmatrix} CD & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 63 & 63 & 63 & 63 \\ 7C & 63 & 63 & 63 \end{bmatrix}$$

c) MixColumns

MixColumns operates column-wise.

Only **first column differs**, so we show full result:

After MixColumns — P_1

$$\begin{bmatrix} 7C & 63 & 63 & 63 \\ 7C & 63 & 63 & 63 \\ 42 & 63 & 63 & 63 \\ 5D & 63 & 63 & 63 \end{bmatrix}$$

After MixColumns — P_2

$$\begin{bmatrix} 3B & 63 & 63 & 63 \\ D2 & 63 & 63 & 63 \\ EC & 63 & 63 & 63 \\ B4 & 63 & 63 & 63 \end{bmatrix}$$

d) AddRoundKey (Key = all zeros)

Since key = 0, state remains unchanged:

Final State — P_1

$$\begin{bmatrix} 7C & 63 & 63 & 63 \\ 7C & 63 & 63 & 63 \\ 42 & 63 & 63 & 63 \\ 5D & 63 & 63 & 63 \end{bmatrix}$$

Final State — P_2

$$\begin{bmatrix} 3B & 63 & 63 & 63 \\ D2 & 63 & 63 & 63 \\ EC & 63 & 63 & 63 \\ B4 & 63 & 63 & 63 \end{bmatrix}$$

2 Find the results of the following using Fermat's little theorem: **5**

a. 456^{17} mod 17
b. 145^{102} mod 101
c. 27^{-1} mod 41
d. 70^{-1} mod 101

Note that all moduli are primes.

Answer:

a) $(456^{17} \bmod 17) = (456 \bmod 17) = 14 \bmod 17$

b) $(145^{102} \bmod 101) = [(145^{101} \bmod 101) \times (145 \bmod 101)] \bmod 101$
$= [145 \times 145] \bmod 101 = [44 \times 44] \bmod 101 = 17 \bmod 101$

c) $27^{-1} \bmod 41 = 27^{41-2} \bmod 41 = 27^{39} \bmod 41 = 38 \bmod 41$

d) $70^{-1} \bmod 101 = 70^{101-2} \bmod 101 = 70^{99} \bmod 101 = 13 \bmod 101$

3 Find the results of the following using Euler's theorem: **5**

a. 12^{-1} mod 77
b. 16^{-1} mod 323
c. 20^{-1} mod 403
d. 44^{-1} mod 667

Note that 77 = 7 × 11, 323 = 17 × 19, 403 = 31 × 13, and 667 = 23 × 29.

Answer:

a) $12^{-1} \bmod 77 = 12^{\varphi(77)-1} \bmod 77 = 12^{59} \bmod 77 = 45 \bmod 77$

b) $16^{-1} \bmod 323 = 16^{\varphi(323)-1} \bmod 323 = 16^{287} \bmod 323 = 101 \bmod 323$

c) $20^{-1} \bmod 403 = 20^{\varphi(403)-1} \bmod 403 = 20^{359} \bmod 403 = 262 \bmod 403$

	d) $44^{-1} \bmod 667 = 44^{\varphi(667)-1} \bmod 667 = 44^{615} \bmod 667 = 379 \bmod 667$	
4	Determine how many of the following integers pass the Miller-Rabin primality test: 109, 201, 271, 341, 349. Use base 2. Answer: • $n = 109 \rightarrow 109 - 1 = 27 \times 2^2 \rightarrow m = 27$ and $k = 2$. **Pre-loop** $\rightarrow$ **T** $= 2^{27}$ **mod 109 = 33** $k = 1 \rightarrow$ **T** $=$ **T**^{2} **mod 109 = 108 mod 109 = (−1) mod 109** **Loop is broken because T = −1 → Pseudoprime (109 is actually a prime)** • $n = 201 \rightarrow 201 - 1 = 25 \times 2^3 \rightarrow m = 25$ and $k = 3$. **Pre-loop** $\rightarrow$ **T** $= 2^{25}$ **mod 201 = 95** $k = 1 \rightarrow$ **T** $=$ **T**^{2} **mod 201 = 181 mod 201** $k = 2 \rightarrow$ **T** $=$ **T**^{2} **mod 201 = 199 mod 201** **Loop is terminated → Composite (201 = 3 × 67)** • $n = 271 \rightarrow 271 - 1 = 135 \times 2^1 \rightarrow m = 135$ and $k = 1$. **Pre-loop** $\rightarrow$ **T** $= 2^{135}$ **mod 271 = 1 mod 271** **T = +1 in the initialization step → Pseudoprime (271 is actually a prime)** • $n = 341 \rightarrow 341 - 1 = 85 \times 2^2 \rightarrow m = 85$ and $k = 2$. **Pre-loop** $\rightarrow$ **T** $= 2^{85}$ **mod 341 = 32** $k = 1 \rightarrow$ **T** $=$ **T**^{2} **mod 341 = 1 mod 341** **Loop is broken because T = +1 → Composite (341 = 11 × 31)** • $n = 349 \rightarrow 349 - 1 = 87 \times 2^2 \rightarrow m = 87$ and $k = 2$. **Pre-loop** $\rightarrow$ **T** $= 2^{87}$ **mod 349 = 213** $k = 1 \rightarrow$ **T** $=$ **T**^{2} **mod 349 = 348 mod 349 = (−1) mod 349** **Loop is broken because T = −1 → Pseudoprime (349 is actually a prime)**	10
5	Find the value of x for the following sets of congruence using the Chinese remainder theorem. $x \equiv 2 \bmod 7$, and $x \equiv 3 \bmod 9$ Answer: $a_1 = 2$ $m_1 = 7$ $a_2 = 3$ $m_2 = 9 \rightarrow M = 63$ $M_1 = 9$ $M_1^{-1} = 9^{-1} \bmod 7 = 4$; $M_2 = 7$ $M_2^{-1} = 7^{-1} \bmod 9 = 4$ $x = (2 \times 9 \times 4 + 3 \times 7 \times 4) \bmod 63 = 30$	5
6	Find the result of the following using the square-and-multiply method. $21^{24} \bmod 8$	5

	Answer: $y = 21^{24} \bmod 8 \rightarrow a = 21, x = 24 = 11000_2$. Shaded areas represent no multiplications. All calculations are in modulo 8. The answer is $\boldsymbol{y = 1}$. <table><tr><td>a's</td><td>x_i</td><td>$y = 1 \bmod 8$</td></tr><tr><td>$a^1 = 21 \bmod 8 = 5 \bmod 8$</td><td>0</td><td>$y = 1 \bmod 8$</td></tr><tr><td>$a^2 = 25 \bmod 8 = 1 \bmod 8$</td><td>0</td><td>$y = 1 \bmod 8$</td></tr><tr><td>$a^4 = 1 \bmod 8$</td><td>0</td><td>$y = 1 \bmod 8$</td></tr><tr><td>$a^8 = 1 \bmod 8$</td><td>1</td><td>$y = 1 \times 1 \bmod 8 = 1 \bmod 8$</td></tr><tr><td>$a^{16} = 1 \bmod 8$</td><td>1</td><td>$y = 1 \times 1 \bmod 8 = 1 \bmod 8$</td></tr></table> The table shows that we can stop whenever a^{x_i} becomes 1.	
7	In RSA, given n = 12091 and e = 13: a. Encrypt the message P = "THIS" using the 00−25 encoding scheme, where 00 = "A" and so on. Use different blocks to make P < n. b. Decrypt the ciphertext to find the original message. Answer: Although in real life Alice cannot check to see if n and e are properly selected by Bob, in this toy problem she can. The value of n is proper because $n = 107 \times 113$ (two primes). The value of e is also proper because e, which is 13 and $\varphi(n)$, which is 11872, are coprime. **a.** The plaintext is: **19070818.** For encryption, we create 4-digit blocks so that the value of each block be less than n. **P1 = 1907 → C1 = 1907^{13} mod 12091 = 10614** **P2 = 0818 → C2 = 0818^{13} mod 12091 = 7787** **b.** In this toy problem, we can easily find $d = e^{-1} \bmod \varphi(n) = 3653$. Each block can be decrypted as **C1 = 10614 → P1 = 10614^{3653} mod 12091 = 1907** **C2 = 7787 → P2 = 7787^{3653} mod 12091 = 0818** The plaintext is:19070818 or "THIS".	10
8	Using the Rabin cryptosystem with p = 47 and q = 11: a. Encrypt P = 17 to find the ciphertext. b. Use the Chinese remainder theorem to find four possible plaintexts. Answer: We first find $n = p \times q = 47 \times 11 = 517$. **a.** C = P^2 mod 517 = 172 mod 517 = 289.	10

	b. $a_1 = +\, C^{(p+1)/4} \bmod p = +\, 289^{12} \bmod 47 = +\, 17$ and $a_2 = -17$. $b_1 = +\, C^{(q+1)/4} \bmod q = +\, 289^{3} \bmod 11 = +\, 5$ and $b_2 = -\, 5$. We have the following four sets, each of two equations to solve: Chinese Remainder ($a_1 \equiv +17 \pmod{47}$, $b_1 \equiv +5 \pmod{11}$) → P1 = 346 Chinese Remainder ($a_1 \equiv +\, 17 \pmod{47}$, $b_2 \equiv -5 \pmod{11}$) → P2 = 17 Chinese Remainder ($a_2 \equiv -17 \pmod{47}$, $b_1 \equiv +5 \pmod{11}$) → P3 = 500 Chinese Remainder ($a_2 \equiv -17 \pmod{47}$, $b_2 \equiv -5 \pmod{11}$) → P4 = 171	
9	In ElGamal cryptosystem, given the prime p = 31: a. Choose an appropriate e_1 and d, then calculate e_2. b. Encrypt the message P = “HELLO”; use 00−25 for encoding. Use different blocks to make P < p. c. Decrypt the ciphertext to obtain the plaintext. Answer: **a.** We choose $e_1 = 3$ (a primitive root of $p = 31$) and $d = 10$. Then we have $e_2 = 3^{10} \bmod 31 = 25$. **b.** The common factor for calculation of C_2’s is $e_2^{\,7} \bmod 31 = 25^7 \bmod 31 = 25$. P = “H” = 07 $C_1 = 3^7 \bmod 31 = 17$ $C_2 = 07 \times 25 \bmod 31 = 20$ → C = (17, 20) P = “E” = 04 $C_1 = 3^7 \bmod 31 = 17$ $C_2 = 04 \times 25 \bmod 31 = 07$ → C = (17, 07) P = “L” = 11 $C_1 = 3^7 \bmod 31 = 17$ $C_2 = 11 \times 25 \bmod 31 = 27$ → C = (17, 27) P = “L” = 11 $C_1 = 3^7 \bmod 31 = 17$ $C_2 = 11 \times 25 \bmod 31 = 27$ → C = (17, 27) P = “O” = 14 $C_1 = 3^7 \bmod 31 = 17$ $C_2 = 14 \times 25 \bmod 31 = 09$ → C = (17, 09) **c.** C = (17, 20) → P = $20 \times (17^{10})^{-1} \bmod 31 = 07$ → “H” C = (17, 07) → P = $07 \times (17^{10})^{-1} \bmod 31 = 04$ → “E” C = (17, 27) → P = $27 \times (17^{10})^{-1} \bmod 31 = 11$ → “L” C = (17, 27) → P = $27 \times (17^{10})^{-1} \bmod 31 = 11$ → “L” C = (17, 09) → P = $09 \times (17^{10})^{-1} \bmod 31 = 14$ → “O”	20
10	a. Explain the underlying mathematical problem on which elliptic curve cryptography is based. b. (Challenge Question) Consider the elliptic curve: $y^2 \equiv x^3 + 2x + 2 \pmod{17}$ Assume a base point G = (5,1) on the curve.	10

Using the given curve and base point, describe the steps of:

- Key generation
- Encryption

(You may use small integer values for private keys to simplify calculations.)

(Instead of detailed solution, you can explain part (b) in simple mathematical language as well)

Answer:

a) Underlying mathematical problem behind ECC

Elliptic Curve Cryptography is based on the **Elliptic Curve Discrete Logarithm Problem (ECDLP).**

On an elliptic curve, we choose a base point G. Then for a secret integer d, we compute

$$Q = dG$$

where dG means adding the point G to itself d times.

This forward operation is easy:

- given d and G, compute $Q = dG$

But the reverse problem is hard:

- given G and Q, find d

That hard reverse problem is the **ECDLP**, and ECC security depends on it.

So the core idea is:

- multiplying a point by a number is easy
- recovering the number from the result is hard

(b)

For the curve

$$y^2 \equiv x^3 + 2x + 2 \pmod{17}$$

with base point $G = (5, 1)$:

Key generation

Choose private key $d_B = 3$.
Compute public key:

$$Q_B = 3G$$

First,

$$2G = (6, 3)$$

Then,

$$3G = (10, 6)$$

So:

- Private key = 3
- Public key = $(10, 6)$

Encryption

Let message point $M = (5, 1)$, and choose random integer $k = 2$.

Compute:

$$C_1 = kG = 2G = (6, 3)$$

$$kQ_B = 2(10, 6) = (16, 13)$$

$$C_2 = M + kQ_B = (5, 1) + (16, 13) = (0, 6)$$

Therefore ciphertext is:

$$\boxed{(C_1, C_2) = ((6, 3), (0, 6))}$$